

DZHUSALY, KazakhstanWMO: **359530**Lat: **45.510N**Lon: **64.090E**Elev: **103**StdP: **100.09**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.2	-22.3	-28.1	0.3	-24.8	-25.2	0.4	-22.1	21.1	-14.5	17.4	-11.6	3.9	50	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	38.9	19.5	37.2	18.9	35.7	18.5	21.2	34.4	20.2	33.5	19.4	32.8	4.9	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.5	25.3	15.4	11.1	25.4	13.8	10.0	25.7	61.9	34.3	58.4	33.7	55.8	32.5	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.2	12.9	11.5	DB	-27.0	41.9	4.0	1.5	-29.9	43.0	-32.2	43.8	-34.5	44.7	-37.4	45.7	(4)
(5)			WB	-27.0	22.8	3.9	1.9	-29.9	24.2	-32.2	25.3	-34.4	26.4	-37.2	27.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-9.1	-6.5	3.8	12.9	20.0	25.5	27.4	25.9	18.7	9.6	1.0	-6.0	(6)
(7)		DBStd	13.92	6.81	7.63	5.93	5.27	4.01	3.74	3.33	3.81	4.67	5.03	5.76	6.07	(7)
(8)		HDD10.0	2129	591	461	205	32	1	0	0	2	69	273	497		(8)
(9)		HDD18.3	3797	849	694	452	172	28	2	0	2	51	273	519	755	(9)
(10)		CDD10.0	2263	0	0	12	119	312	467	538	493	262	57	3	0	(10)
(11)		CDD18.3	888	0	0	0	10	81	218	280	237	61	2	0	0	(11)
(12)		CDH23.3	12283	0	0	9	171	1102	3062	3998	3147	747	47	0	0	(12)
(13)		CDH26.7	6603	0	0	2	46	457	1696	2318	1766	308	10	0	0	(13)
(14)	Wind	WSAvg	4.6	4.6	4.6	5.0	5.1	5.0	4.7	4.5	4.6	4.2	4.0	3.9	4.4	(14)
(15)	Precipitation	PrecAvg	136	14	13	14	16	15	10	6	5	3	10	16	15	(15)
(16)		PrecMax	205	33	28	37	44	41	44	54	40	24	57	36	42	(16)
(17)		PrecMin	73	3	3	2	3	3	1	0	0	0	0	3	1	(17)
(18)		PrecStd	35	8	7	8	12	11	9	10	8	5	12	10	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	11.1	23.7	31.0	35.7	40.0	41.8	40.1	36.4	27.8	18.1	7.0	(19)
(20)			MCWB	4.0	7.6	12.7	15.9	17.4	19.6	21.7	19.3	17.8	14.0	11.5	4.1	(20)
(21)		2%	DB	2.7	6.7	18.7	27.7	33.3	38.0	39.3	38.1	32.8	24.2	14.0	4.5	(21)
(22)			MCWB	1.8	3.8	10.5	14.5	16.7	19.0	20.0	18.9	16.0	12.1	8.8	3.0	(22)
(23)		5%	DB	1.2	4.1	15.4	24.9	31.4	36.2	37.3	36.5	30.1	21.4	11.3	2.7	(23)
(24)			MCWB	0.4	2.2	8.8	13.0	15.9	18.3	19.3	18.4	15.3	11.1	7.3	1.6	(24)
(25)		10%	DB	-0.1	2.2	12.4	22.4	29.3	34.5	35.6	34.7	27.5	18.6	8.7	1.1	(25)
(26)			MCWB	-0.8	0.7	7.5	12.0	15.4	17.7	19.0	18.0	14.3	10.0	5.6	0.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	7.9	13.3	17.3	20.1	21.9	23.7	22.0	18.6	14.8	12.7	5.3	(27)
(28)			MCDWB	5.4	10.9	22.4	27.5	31.5	34.6	34.9	33.3	33.1	25.5	16.4	6.4	(28)
(29)		2%	WB	2.0	4.3	11.1	15.7	18.1	20.1	21.6	20.5	17.0	13.1	9.6	3.2	(29)
(30)			MCDWB	2.6	6.0	17.0	24.6	29.1	33.4	35.3	34.2	30.6	21.7	13.1	4.3	(30)
(31)		5%	WB	0.5	2.2	9.5	14.1	17.0	19.3	20.5	19.5	15.9	11.7	7.5	1.6	(31)
(32)			MCDWB	1.1	3.9	14.7	23.0	28.4	32.7	33.8	33.6	28.5	20.0	10.6	2.5	(32)
(33)		10%	WB	-0.8	0.8	7.8	12.6	16.0	18.5	19.7	18.6	14.7	10.4	5.8	0.3	(33)
(34)			MCDWB	0.0	2.0	12.1	20.5	27.4	32.0	32.9	32.7	26.6	17.7	8.6	1.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.6	8.9	10.9	12.9	13.9	14.2	13.6	13.9	13.9	13.1	9.0	7.1	(35)
(36)			MCDBR	6.4	9.9	15.6	16.5	16.5	16.0	15.3	15.9	16.5	16.9	13.0	7.4	(36)
(37)			MCWBR	5.7	7.4	8.8	7.4	6.4	5.7	5.5	5.7	6.8	8.7	8.6	6.1	(37)
(38)		5% WB	MCDBR	5.9	9.1	14.4	14.7	14.7	14.5	13.8	14.2	15.3	15.5	11.6	6.7	(38)
(39)			MCWBR	5.4	7.1	8.8	7.5	6.7	5.8	5.7	5.8	7.0	8.6	8.0	5.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.341	0.367	0.428	0.469	0.471	0.450	0.432	0.417	0.383	0.387	0.370	0.349	(40)	
(41)		taud	2.074	2.041	1.987	1.920	1.969	2.085	2.162	2.180	2.220	2.124	2.126	2.090	(41)	
(42)		Ebn at Noon	718	779	783	789	803	822	832	830	827	750	680	658	(42)	
(43)		Edh at Noon	100	126	154	179	176	158	145	137	120	114	95	89	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.37	2.51	3.81	5.36	6.82	7.37	7.18	6.50	5.09	3.20	1.78	1.13	(44)	
(45)		RadStd	0.29	0.37	0.31	0.38	0.38	0.36	0.31	0.24	0.16	0.19	0.17	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-1.94	N/A	N/A	N/A	N/A	(46)
(47) Regional (3 neighbors)	N/A	N/A	N/A	+0.88	N/A	-0.84	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page